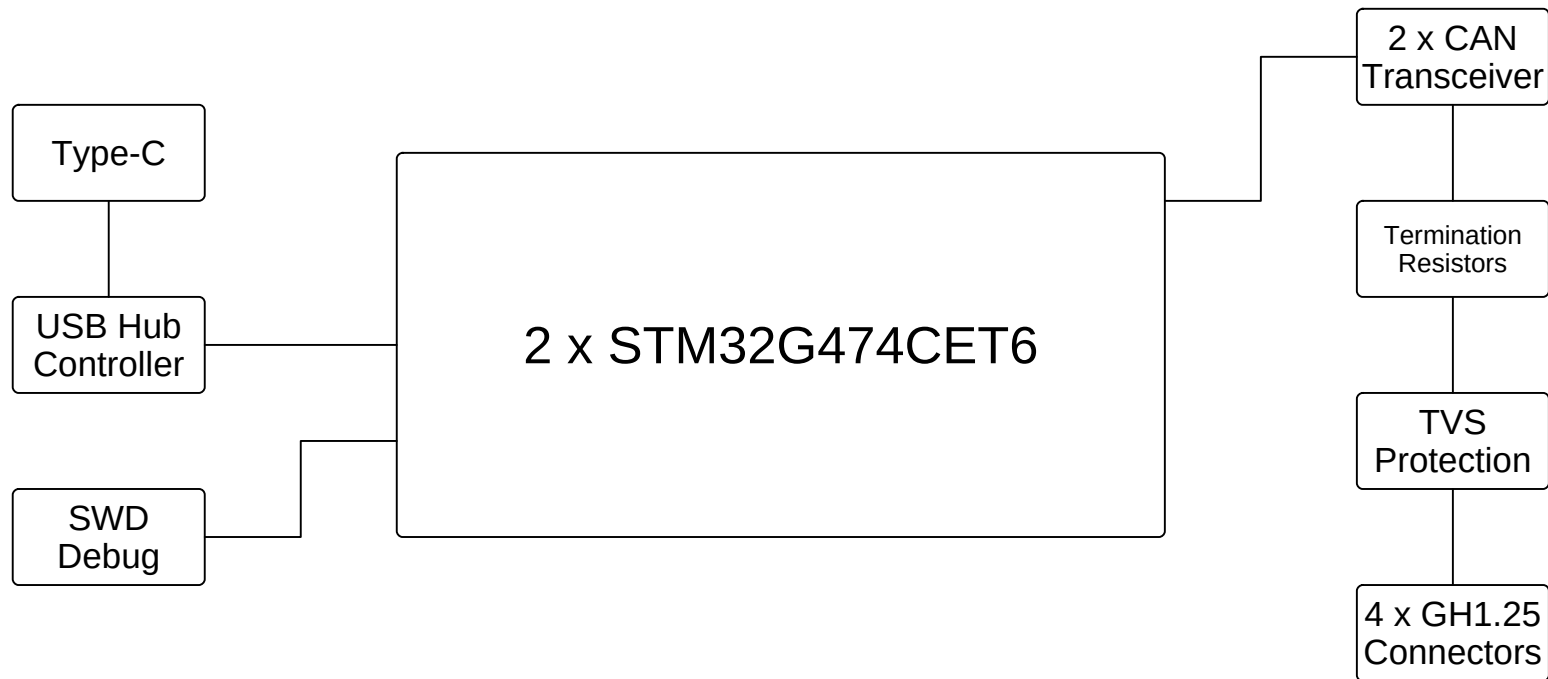
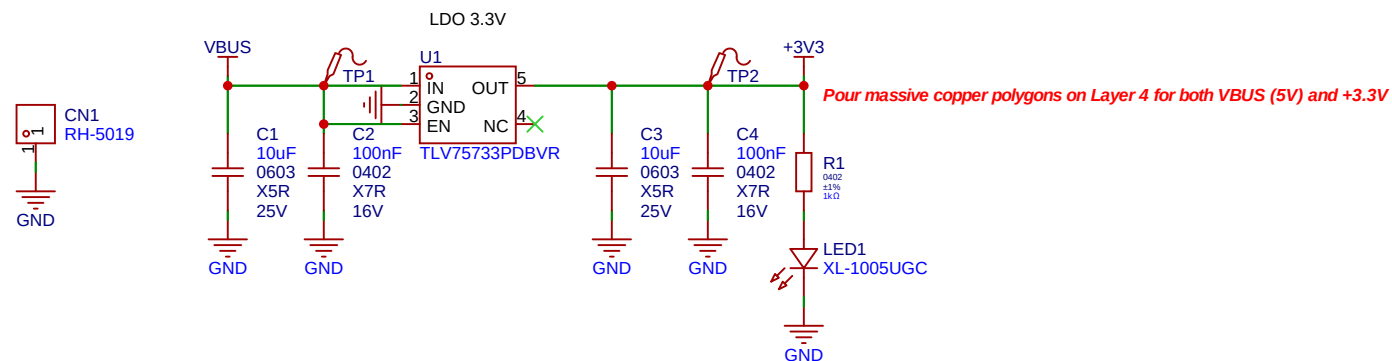


Date	Version Number	Details
July 25th 2026	V1.0	Finished on August 11th 2026

Schematic	Schematic		Create at	2026-07-25
			Update at	2026-08-11
Board	Board_1_0		Page	Version
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 1 Total 8
		V0.0	A4	VNX Robotics

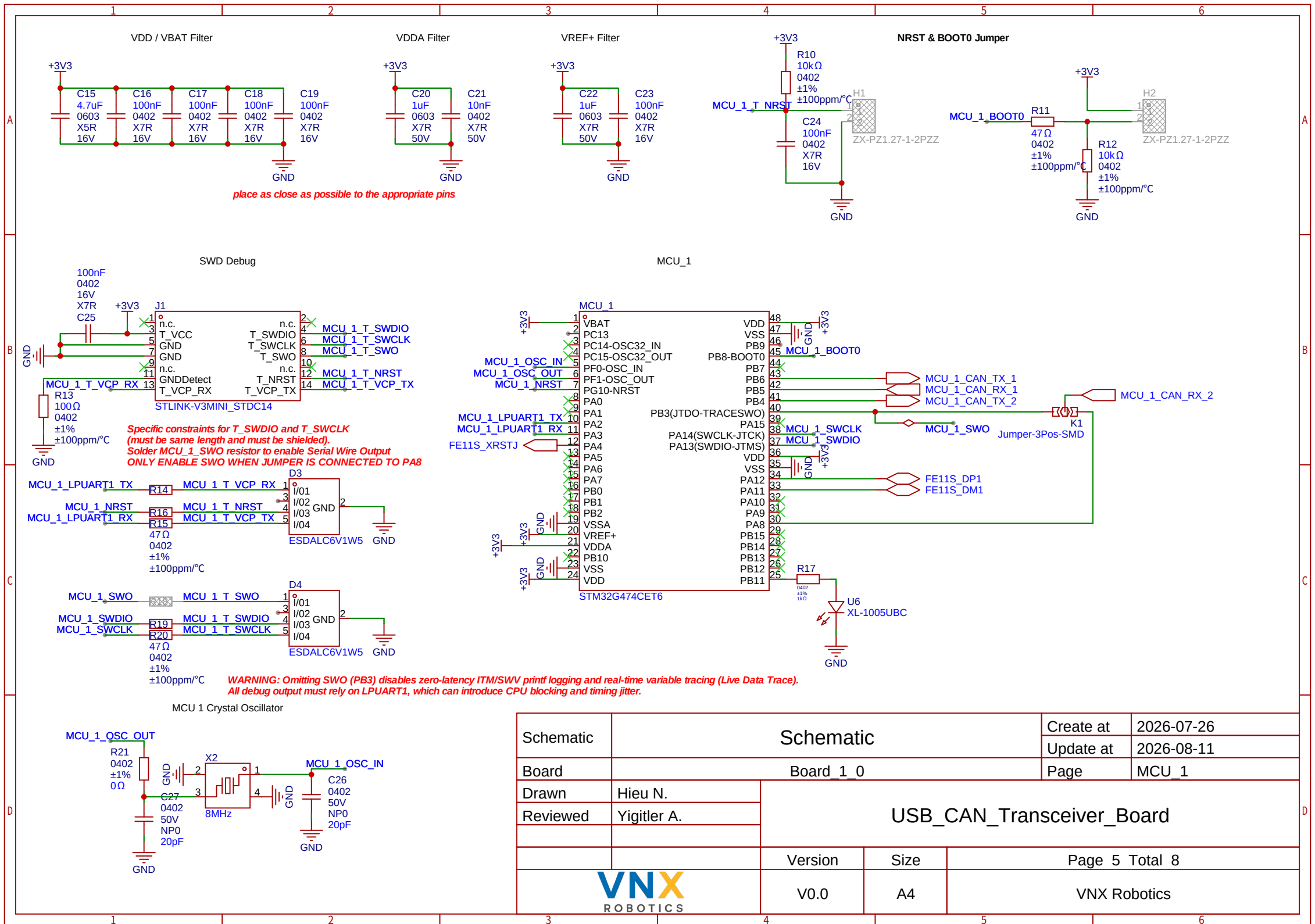


Schematic	Schematic			Create at	2026-07-25
Board	Board_1_0			Update at	2026-08-08
Drawn	Hieu N.	USB_CAN_Transceiver_Board			
Reviewed	Yigitler A.				
		Version	Size	Page 2 Total 8	
		V0.0	A4	VN X Robotics	

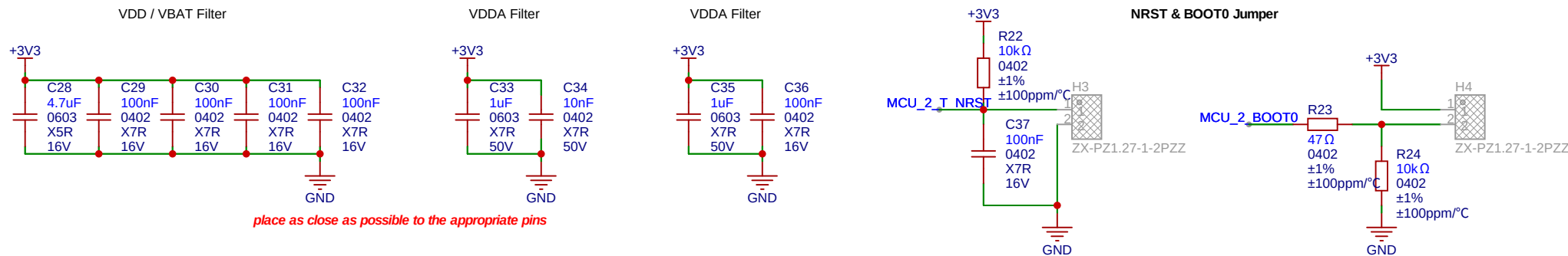


Component	Current Draw (Approx.)
FE1.1S-BSOP28BCN	70 mA @ 5V
STM32G474CET6 (x2)	35 mA each (running at 170 MHz) @ 3.3V
Can Transceiver Logic	20 mA @ 3.3V
LEDs	10mA @ 3.3V
Can Transceiver	@ 5V

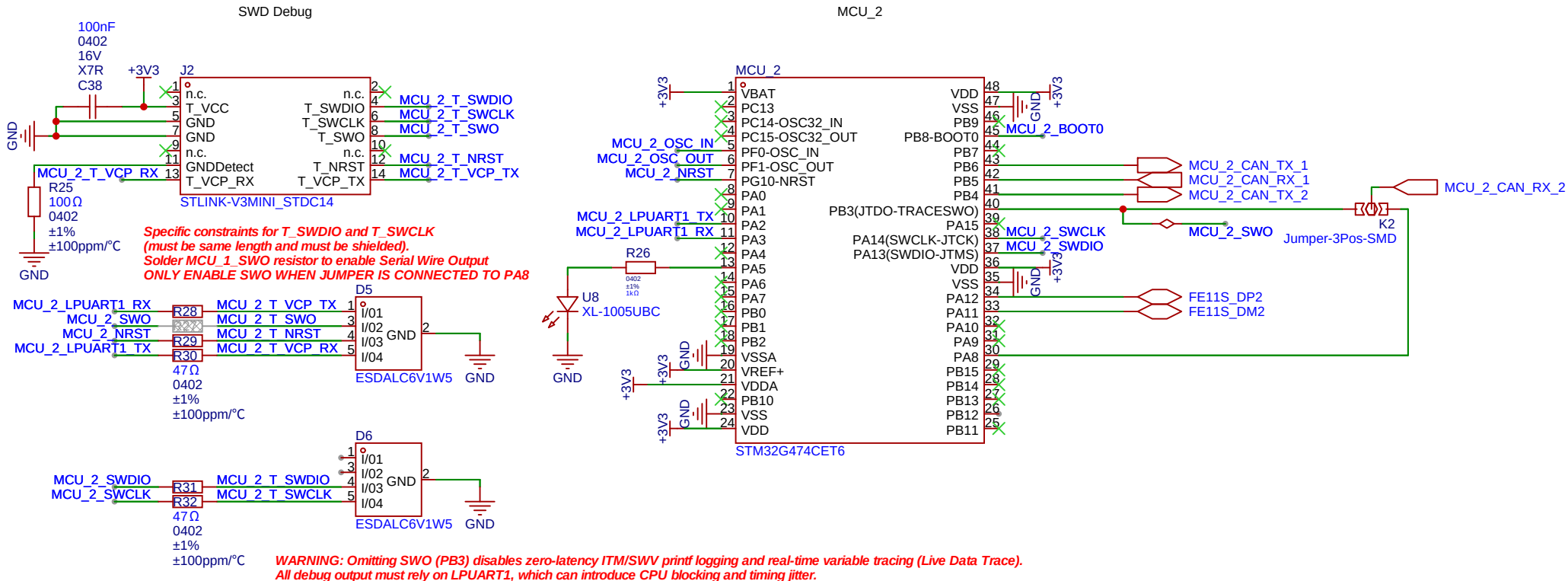
Schematic	Schematic		Create at	2026-07-26
			Update at	2026-08-11
Board	Board_1_0		Page	Power
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 3 Total 8
VN X ROBOTICS		V0.0	A4	VNX Robotics



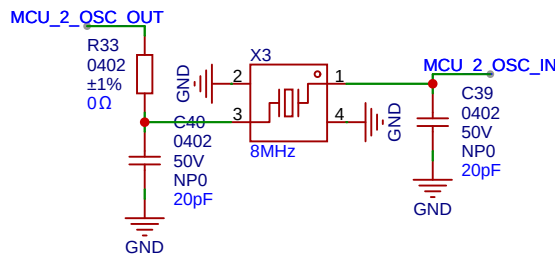
Short & symmetrical OSC traces. Isolated Local GND plane. No signal/RF crossovers.



place as close as possible to the appropriate pins



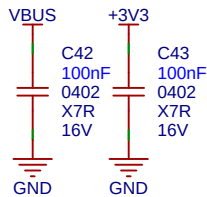
MCU 1 Crystal Oscillator



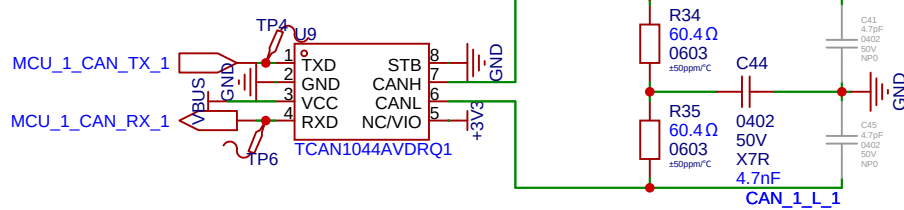
Schematic	Schematic		Create at	2026-08-05
			Update at	2026-08-11
Board	Board_1_0		Page	MCU_2
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 6 Total 8
VN X ROBOTICS		V0.0	A4	VNX Robotics

Short & symmetrical OSC traces. Isolated Local GND plane. No signal/RF crossovers.

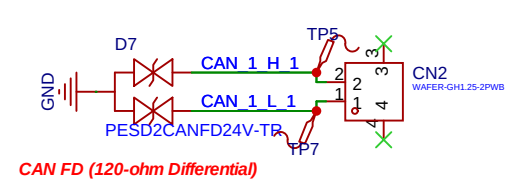
VCC / VIO Filter



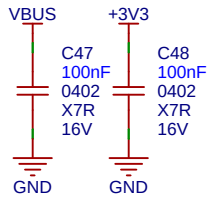
CAN Transceiver 1



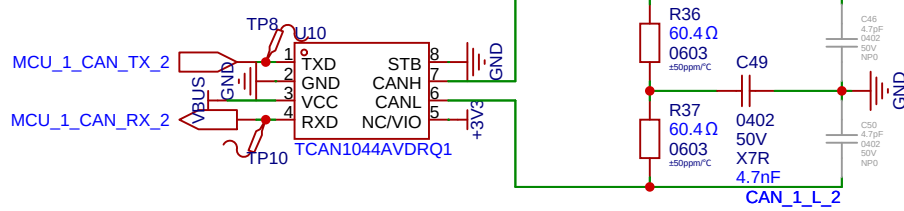
CAN Transceiver 1 Protection



VCC / VIO Filter

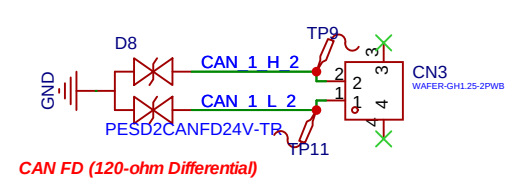


CAN Transceiver 2

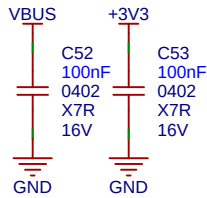


Place the split termination networks halfway between the transceiver pins and the GH1.25 headers

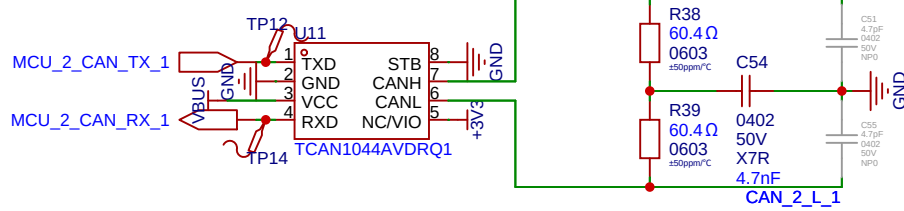
CAN Transceiver 2 Protection



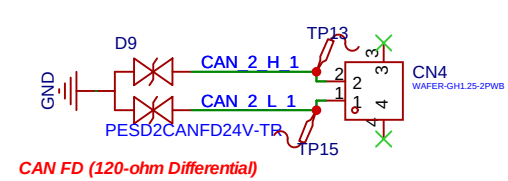
VCC / VIO Filter



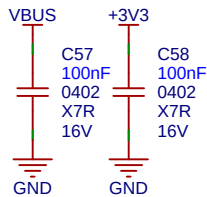
CAN Transceiver 3



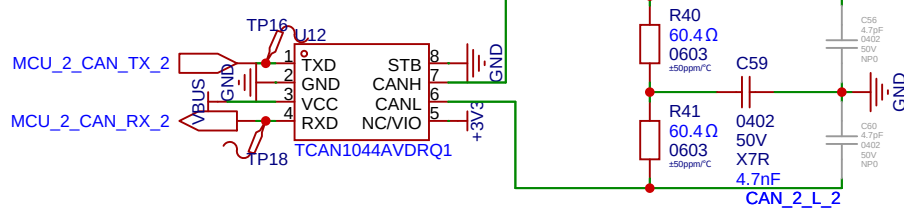
CAN Transceiver 2 Protection



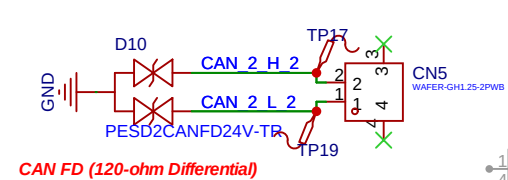
VCC / VIO Filter



CAN Transceiver 4



CAN Transceiver 2 Protection

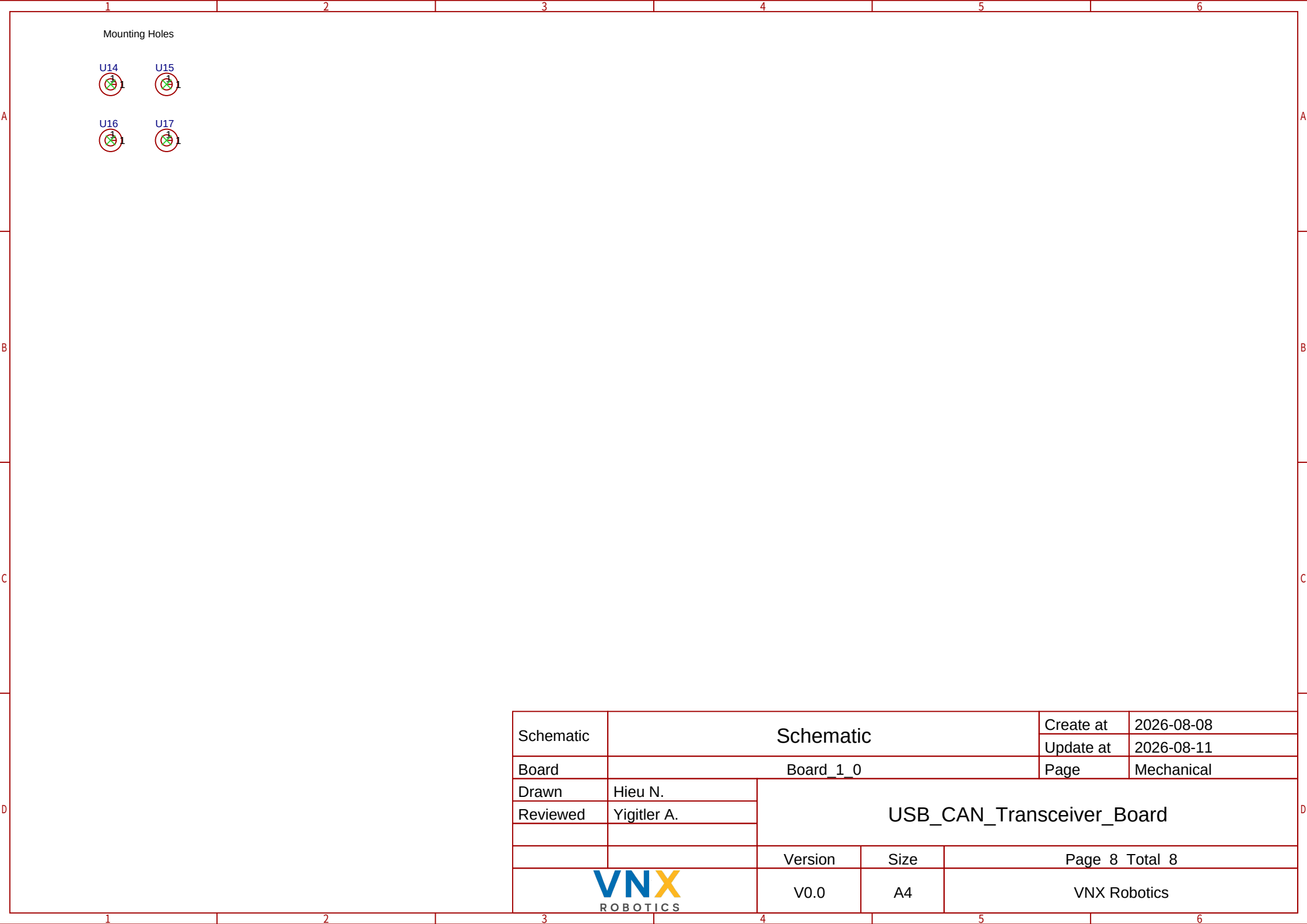


UMI's solution



Specification / Parameter	TCAN3414DRBR	TCAN1044AVDRQ1
Main Supply Voltage (VCC)	3.0V to 3.6V (Nominal 3.3V)	4.5V to 5.5V (Nominal 5.0V)
Logic Level Shifting (VIO)	No (Logic referenced to VCC)	Yes (VIO supports 1.7V to 5.5V)
Supported Data Rates	Classical CAN & CAN FD up to 5 Mbps (Up to 8 Mbps in simple networks)	
Recessive Bus Bias Voltage	Nominal 1.9V	Nominal 2.5V (VCC / 2)
Operating Modes	Normal, Low-Power Standby, Ultra-Low Power Shutdown	Normal, Low-Power Standby (with Remote Wake WUP)
Application	Industrial	Automotive

Schematic		Schematic			Create at	2026-07-26
					Update at	2026-08-11
Board		Board_1_0			Page	CAN
Drawn	Hieu N.	USB_CAN_Transceiver_Board				
Reviewed	Yigitler A.					
		Version	Size	Page 7 Total 8		
		V0.0	A4	VN X Robotics		



Schematic	Schematic		Create at	2026-08-08
			Update at	2026-08-11
Board	Board_1_0		Page	Mechanical
Drawn	Hieu N.	USB_CAN_Transceiver_Board		
Reviewed	Yigitler A.			
		Version	Size	Page 8 Total 8
VN X ROBOTICS		V0.0	A4	VNX Robotics